

- 1** On a particular day in 2023, the exchange rate between Singapore dollars (SGD) and United States dollars (USD) was USD 1 = SGD x .

- (a)** Write down an expression, in terms of x , for the USD received in exchange for SGD 10 000.

Answer USD = _____ [1]

In 2022, the rate was such that USD 1 would exchange for SGD 0.030 **less** than in 2023. The difference in exchanging SGD 10 000 in 2023 and SGD 10 000 in 2022, is USD 166.

- (b)** Write down an equation to represent this information and show that it reduces to $83x^2 - 2.49x - 150 = 0$.

Answer

[3]

- (c)** Solve the equation $83x^2 - 2.49x - 150 = 0$, giving your solutions correct to three decimal places.

Answer $x =$ _____ or _____ [3]

- (d)** Find the amount of USD received in exchange for SGD 20 000 in 2022. Give your answer correct to the nearest dollar.

Answer USD = _____ [2]

2 (a) It is given that $sq^2 = r - 3q^2$.

(i) Evaluate s when $q = -3$ and $r = 5$.

(ii) Express q in terms of r and s . Answer $s =$ _____ [1]

Answer $q =$ _____ [2]

(b) (i) Show that $(7n-1)^2 - (n-1)^2$ is a multiple of 12 for all integer values of n .

Answer

[2]

(b) (ii) Simplify $\frac{p-3n+12n^2-4np}{(7n-1)^2-(n-1)^2}$.

Answer _____ [3]
